There is some debate about the opportunity to consider the latter either as algorithms or rather heuristics, but I lean toward including them, as the sequence of steps in those algorithms is deterministic and well defined. In this book we present a few Monte Carlo algorithms, so you will have a chance to form an educated opinion.

But let’s first define a particularly useful class of *problems*, instead, decision problems.

## F.1 Decision Problems

When talking about algorithms that return either `true` or `false`, we fall into the class of problems called *binary decision problems*.

Binary classification algorithms assign one of two labels to the data. The labels can actually be anything, but conceptually this is equivalent to assigning a Boolean label to each point, so we'll use this notation in the rest of the book.

Considering only binary classification is not restrictive since multiclass classifiers can be obtained by combining several binary classifiers.

Even more, one fundamental result in the computer science, and in particular in the field of computability and complexity, is that any optimization problem can be expressed as a decision problem and vice-versa.